

Model 2: Plotting

This file contains the processing code for Model 2. Before running this file, you must have extinction time outputs from running Model 2 for each species for both time periods using Model2.nb. Edit import ('directory') and export ('saveto') locations for your own device.

Figure 5

Code for each subfigure of figure 6. Parts a and b require only T.citricidus (or any other species), but parts c-e require the full dataset.

```
directory = "/IMPORT/";  
saveto = "/EXPORT/";
```

Subfigs a & b

```
In[3]:= cmplx[mod_, arg_] := ExpToTrig[mod Exp[I arg]];

SpecSynFourier[γ_, Nobs_, μ_ : 0, σ_ : 1, seed_ : 0] := Module[{  
  phase, f, fseries, vec},  
  If[seed == 0, SeedRandom[], SeedRandom[seed]];  
  phase = RandomReal[{0, 2 Pi}, Nobs];  
  f = Range[1 / Nobs, 0.5, 1 / Nobs];  
  fseries =  
    Join[{0 + 0 I}, Table[cmplx[1 / f[[i]] ^ (γ / 2), phase[[i]], {i, 1, Length[f]}],  
      Table[cmplx[1 / f[[i]] ^ (γ / 2), -1 * phase[[i]], {i, Length[f] - 1, 1, -1}]];  
  vec = InverseFourier[fseries, FourierParameters → {-1, 1}];  
  Standardize[Re[vec[[1 ;; Nobs]]] * σ + μ];

changeautocorr[X_, γ_, samplesperyear_, seed_ : 0] := Module[{  
  var, Y, fftdata, mean, mmeandevs, adjvar,  
  powersp, int, lm, a, b, n = Length[X], phase, f, vec, Z},  
  mean = Mean[X];  
  var = Variance[X - mean];  
  mmeandevs =  
    Table[Mean[Table[X[[i]], {i, j, n, samplesperyear}]], {j, 1, samplesperyear}];  
  Y = Table[X[[i]] - mmeandevs[[Mod[i, samplesperyear, 1]], {i, 1, n}];  
  adjvar = Variance[Y];  
  fftdata = Fourier[Y, FourierParameters → {-1, 1}];  
  powersp = Table[{i / n, Abs[fftdata[[i + 1]] ^ 2], {i, 1, n / 2 - 1}}];
```

```

lm = LinearModelFit[{Log[#1], Log[#2]} &@@@ powersp, x, x];
If[seed == 0, SeedRandom[], SeedRandom[seed]];
phase = RandomReal[{0, 2 Pi}, n];
f = Range[1/n, 1, 1/n];
vec = InverseFourier[
  Join[{0 + 0 I}, Table[cmplx[1/f[[i]]^( $\gamma/2$ ), phase[[i]]], {i, 1, Length[f]}],
  Table[cmplx[1/f[[i]]^( $\gamma/2$ ), -1*phase[[i]]], {i, Length[f] - 1, 1, -1}]],
  FourierParameters -> {-1, 1}];
Z = Standardize[Re[vec]] * Sqrt[adjvar];
{{lm, var, adjvar},
 Table[Z[[i]] + mmeandevs[Mod[i, samplesperyear, 1]], {i, 1, n}]}
]

Mimic1[x_, z_] := Module[{y},
  y = x;
  Do [y[[Ordering[z, i][[i]]] = x[[i]], {i, 1, Length[x]}];
  y];

getParameters[species_String] :=
Module[{row}, row = SelectFirst[speciesData, #[[1]] == species &];
  If[row === Missing["NotFound"], "Species not found", Rest[row]]]

species = "T.citricidus"; (*select species. watch out for spelling!*)
when = "1994-01-01to2003-12-31";
(*1994-01-01to2003-12-31 OR 2014-01-01to2023-12-31*)

TPCdata = Import[directory <> "speciesparams.xlsx"];
speciesData = TPCdata[[1, 2 ;;]];

parameters = getParameters[species];
CTmax = parameters[[1]];
Topt = parameters[[2]];
 $\sigma_p$  = parameters[[3]];
lat = ToString[parameters[[4]];
long = ToString[parameters[[5]];
name = parameters[[14]];

data = Import[directory <> "VisualCrossingData/" <>
  species <> "_" <> lat <> "," <> long <> "_" <> when <> ".csv", "CSV"];
temps = Flatten[data[[2 ;;, {2, 3}]]];
pairedTemps = Partition[temps, 2];
meanTemps = Mean /@ pairedTemps;

 $\gamma$  = 1;

```

```

X = changeautocorr[meanTemps, 7, 365, 0];
W = SortBy[pairedTemps, Mean[{#[[1]], #[[2]]}] &];
Z = Mimic1[W, X[[2]]];
Z = Flatten[Z];
Z2 = Select[Z, # ≥ -15 &]; (*excluding a couple of cold outliers for visibility -
    they are not excluded from simulations*)

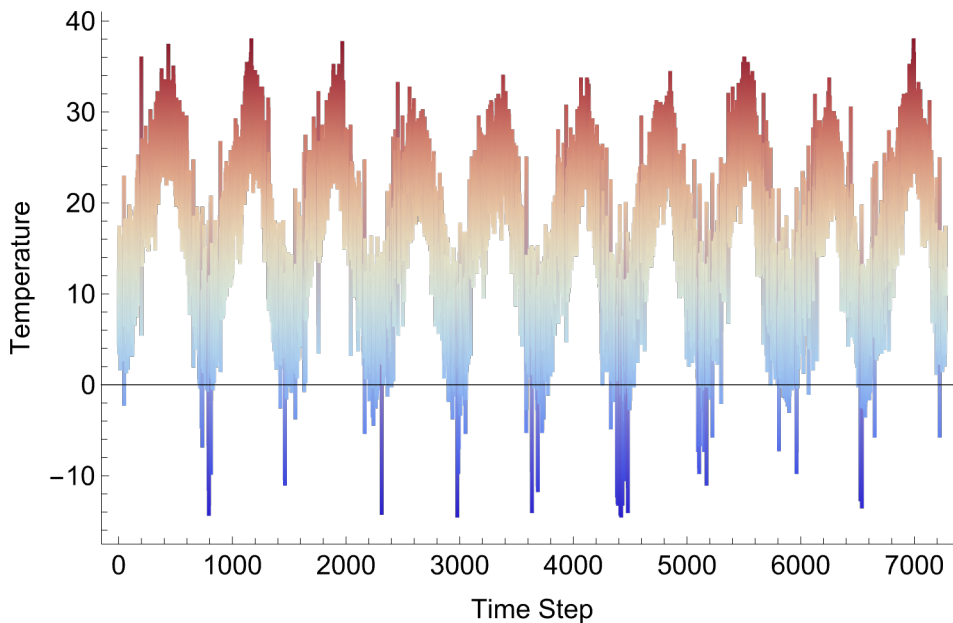
tempsubplot = ListLinePlot[Z2, ColorFunction → ColorData["ThermometerColors"],
    Frame → {True, True, False, False}, FrameStyle → 14,
    FrameLabel → {"Time Step", "Temperature"},
    LabelStyle → Directive[Black], ImageSize → 500]

rD[T_, Topt_, σp_, CTmax_] :=
    If[T ≤ Topt, Exp[-((T - Topt) / (2 σp))^2], 1 - ((T - Topt) / (Topt - CTmax))^2];
TcTPC = Plot[rD[T, Topt, σp, CTmax], {T, 3, 35}, PlotRange → {{3, 34}, {-0.2, 1.1}},
    AxesOrigin → {-10, 0}, PlotStyle → Darker[Purple], Filling → Axis,
    Frame → {True, False, False, True}, FrameStyle → 18,
    FrameLabel → {"Temperature, T (°C)", None, None, "Normalized Fitness r"},
    LabelStyle → Directive[Black], FrameTicks → {{10, 20, 30}, {0, 0.5, 1}},
    Epilog → {Style[Text[Style[ToString[name], FontWeight → Bold, FontSlant → Italic],
        {10, .6}], 18]], ImageSize → 500]

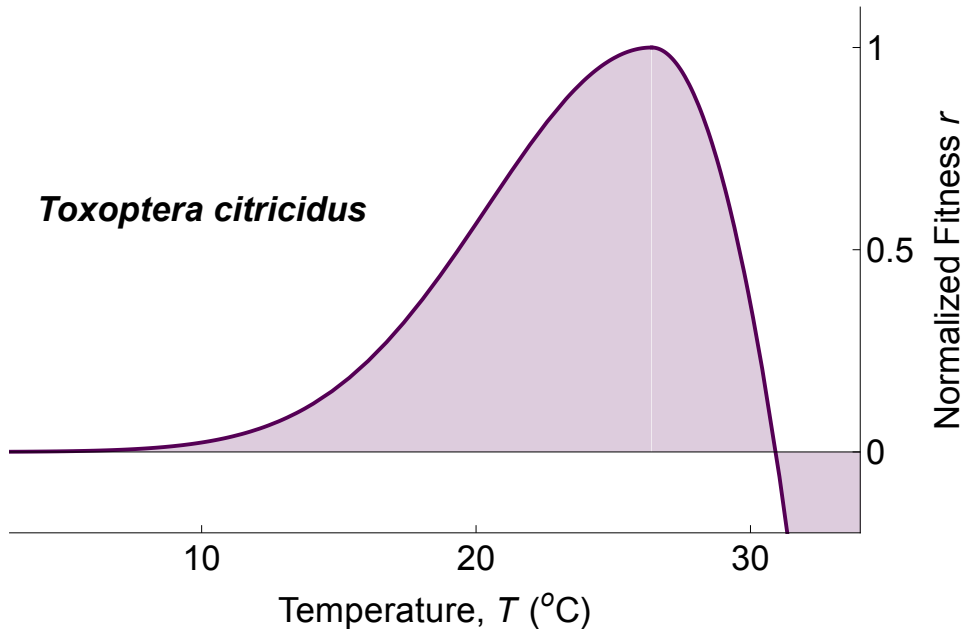
Export[saveto <> "tempsubplot.pdf", tempsubplot, ImageResolution → 1000];
Export[saveto <> "TcitracidusTPC.pdf", TcTPC, ImageResolution → 1000];

```

Out[29]=



Out[31]=



Dot plot summary of results

These figures require .csvs of point locations.

```
In[34]:= usefuldata = Import[directory <> "usefulpoints.csv"];
speciesData1 = Rest[usefuldata];
points1 = speciesData1 /. {species_, hGamma_, hPExt_, pGamma_, pPExt_} =>
  {species, {{hGamma, hPExt}, {pGamma, pPExt}}};

uselessdata = Import[directory <> "uselesspoints.csv"];
speciesData2 = Rest[uselessdata];
points2 = speciesData2 /. {species_, hGamma_, hPExt_, pGamma_, pPExt_} =>
  {species, {{hGamma, hPExt}, {pGamma, pPExt}}};

pext = Graphics[Table[{PointSize[Large], LightGray,
  Point[points2[[i, 2, 1]], Gray, Point[points2[[i, 2, 2]], Black,
  Dotted, Line[points2[[i, 2]]], {i, 1, Length[points2]}},
  Table[{PointSize[Large], Lighter[Lighter[Lighter[Red]]],
  Point[points1[[i, 2, 1]], Red, Point[points1[[i, 2, 2]],
  Black, Dashed, Line[points1[[i, 2]]],
  {i, 1, Length[points1]}},
  Axes -> True, AxesLabel -> {"Gamma", "P(ext)"}, PlotRange ->
  {{0.2, 0.9}, {-0.03, 1.02}}, Frame -> {True, True, False, False}, FrameStyle -> 14,
  FrameLabel -> {(*"Spectral Exponent  $\gamma$ "*)None, "Probability of Extinction"},
  LabelStyle -> Directive[Black],
  GridLines -> {Range[0.2, 0.9, 0.1], Range[0, 1, 0.2]},
```

```

FrameTicks → {None, Automatic}, ImageSize → 500, AspectRatio →  $\frac{13}{23}$ ];

exttimedata = Import[directory <> "fullextinctpoints.csv"];
speciesData3 = Rest[exttimedata];
points3 = speciesData3 /. {species_, hGamma_, hPExt_, pGamma_, pPExt_} =>
  {species, {{hGamma, hPExt}, {pGamma, pPExt}}};

exttime =
Graphics[Table[{PointSize[Large], If[i ≤ 3, Lighter[Lighter[Lighter[Orange]]],
  Lighter[Lighter[Lighter[Purple]]]}, Point[points3[[i, 2, 1]],
  If[i ≤ 3, Orange, Purple], Point[points3[[i, 2, 2]], Black,
  Dashed, Line[points3[[i, 2]]], {i, 1, Length[points3]}]],
Axes → True, AxesLabel → {"Gamma", "P(ext)"}, PlotRange →
  {{0.2, 0.9}, {-50, 1200}}, Frame → {True, True, False, False}, FrameStyle → 14,
FrameLabel → {(*"Spectral Exponent  $\gamma$ "*)None, "Average Time to Extinction"},
LabelStyle → Directive[Black],
GridLines → {Range[0.2, 0.9, 0.1], Range[0, 1200, 200]},
FrameTicks → {None, Automatic}, ImageSize → 500, AspectRatio →  $\frac{13}{23}$ ];

minpopdata = Import[directory <> "fullpersistpoints.csv"];
speciesData4 = Rest[minpopdata];
points4 = speciesData4 /. {species_, hGamma_, hPExt_, pGamma_, pPExt_} =>
  {species, {{hGamma, hPExt}, {pGamma, pPExt}}};

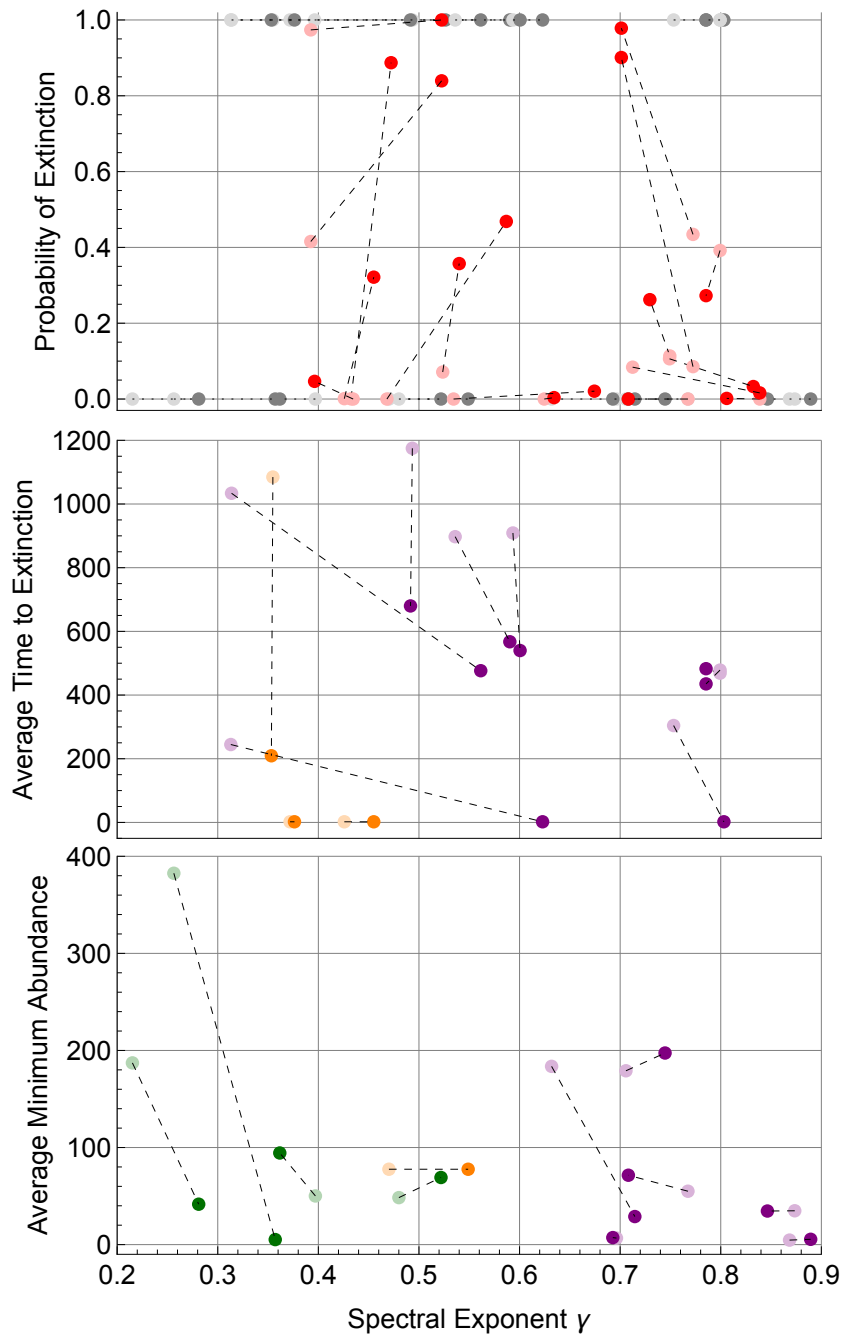
minpop =
Graphics[Table[{PointSize[Large], If[i == 1, Lighter[Lighter[Lighter[Orange]]],
  If[i ≤ 5, Lighter[Lighter[Lighter[Darker[Darker[Green]]]]],
  Lighter[Lighter[Lighter[Purple]]]}, Point[points4[[i, 2, 1]], If[i == 1,
  Orange, If[i ≤ 5, Darker[Darker[Green]], Purple]], Point[points4[[i, 2, 2]],
  Black, Dashed, Line[points4[[i, 2]]], {i, 1, Length[points4]}]],
Axes → True, PlotRange → {{0.2, 0.9}, {-10, 400}},
Frame → {True, True, False, False}, FrameStyle → 14,
FrameLabel → {"Spectral Exponent  $\gamma$ ", "Average Minimum Abundance"},
LabelStyle → Directive[Black],
GridLines → {Range[0.2, 0.9, 0.1], Range[0, 400, 100]},
FrameTicks → {Automatic, Automatic}, ImageSize → 500, AspectRatio →  $\frac{13}{23}$ ];

pointsubplots = GraphicsGrid[{{pext}, {exttime}, {minpop}}, Spacings → -5]

Export[saveto <> "pointsubplots.pdf", pointsubplots, ImageResolution → 1000];

```

Out[49]=



Extended Figures

This code recreates the extended figure plot (containing TPC, temperature histograms, and extinction curves) for an individual species. Most design elements will update automatically, but 'xloc', 'coordi-
nateloc', and 'legendloc' require manual aesthetic adjustments for some species.

In[293]:=

```
getParameters[species_String] :=
```

```

Module[{row}, row = SelectFirst[speciesData, #[[1]] == species &];
  If[row === Missing["NotFound"], "Species not found", Rest[row]]]
TPCdata = Import[directory <> "speciesparams.xlsx"];
rD[T_, Topt_,  $\sigma$ _, CTmax_] :=
  If[T ≤ Topt, Exp[-((T - Topt) / (2  $\sigma$ )) ^ 2], 1 - ((T - Topt) / (Topt - CTmax)) ^ 2];

species = "D.platanoids";

speciesData = TPCdata[[1, 2 ;;]];
parameters = getParameters[species];
CTmax = parameters[[1]]; Topt = parameters[[2]];  $\sigma$  = parameters[[3]];
lat = ToString[parameters[[4]]; long = ToString[parameters[[5]];
Hgamma = parameters[[6]]; Pgamma = parameters[[7]];
Hmean = parameters[[10]]; Hstdev = parameters[[11]];
Pmean = parameters[[12]]; Pstdev = parameters[[13]];
name = parameters[[14]]; region = parameters[[15]];

Hdata = Import[directory <> "VisualCrossingData/" <> species <> "_" <>
  lat <> "," <> long <> "_" <> "1994-01-01to2003-12-31" <> ".csv", "CSV"];
Htemps = Flatten[Hdata[[2 ;;, {2, 3}]]];
Pdata = Import[directory <> "VisualCrossingData/" <> species <> "_" <>
  lat <> "," <> long <> "_" <> "2014-01-01to2023-12-31" <> ".csv", "CSV"];
Ptemps = Flatten[Pdata[[2 ;;, {2, 3}]]];

(*subfig1: TPC*)
plotStyle = Switch[region, "NHEX",
  Darker[Purple], "TROP", Darker[Darker[Green]], "SHEX", Orange];

xloc = 5; (*2*)
fig1 = Plot[rD[T, Topt,  $\sigma$ , CTmax], {T, -10, 43},
  PlotRange → {{-10, 43}, {-0.2, 1.1}}, AxesOrigin → {-10, 0}, PlotStyle → plotStyle,
  Filling → Axis, Frame → {True, True, False, False}, FrameStyle → 14,
  FrameLabel → {"Temperature, T (°C)", "Normalized Fitness r"}, LabelStyle →
  Directive[Black], FrameTicks → {Automatic, {0, 0.25, 0.5, 0.75, 1}}, Epilog →
  {Style[Text[Style[ToString[name], FontWeight → Bold, FontSlant → Italic],
    {xloc, 1}], 14], Style[Text["Tmax = " <> ToString[CTmax], {xloc, .9}], 14],
  Style[Text["Topt = " <> ToString[Topt], {xloc, .8}], 14],
  Style[Text[" $\sigma$ p = " <> ToString[ $\sigma$ ], {xloc, .7}], 14]}, ImageSize → 500];

(*subfig2: temp histogram*)
coordinateloc = {28, 890}; (*select depending on hist range*)
left = {.18, .8}; right = {.8, .8};
legendloc = right; (*depending on histograms, choose where this fits*)

```

```

fig2 = Histogram[{Htemps, Ptemps}, Frame → {True, True, False, False},
  FrameStyle → 14, FrameLabel → {"Temperature, T (°C)", "Frequency"},
  LabelStyle → Directive[Black], ChartLegends →
    Placed[{"Historic (1994–2003)", "Present (2014–2023)"}, legendloc],
  Epilog → {Style[Text[Style["Coordinates: " <> ToString[lat] <> ", " <>
    ToString[long], FontWeight → Bold], coordinateloc], 14],
    Darker[Lighter[Orange]], Thick, Line[{{Hmean, 0}, {Hmean, 2400}}],
    Darker[Lighter[Blue]], Thick, Line[{{Pmean, 0}, {Pmean, 2400}}],
    Darker[Lighter[Orange]], Dashed,
    Line[{{Hmean + Hstdev, 0}, {Hmean + Hstdev, 2400}}],
    Darker[Lighter[Orange]], Dashed,
    Line[{{Hmean - Hstdev, 0}, {Hmean - Hstdev, 2400}}],
    Darker[Lighter[Blue]], Dashed,
    Line[{{Pmean + Pstdev, 0}, {Pmean + Pstdev, 2400}}],
    Darker[Lighter[Blue]], Dashed,
    Line[{{Pmean - Pstdev, 0}, {Pmean - Pstdev, 2400}}]
  }, ImageSize → 500];

```

```
(*subfig3: P(ext)*)
```

```

extTimeH = Import[directory <> "Model3.3_VCOutputs/model3.3.2_" <>
  species <> "_1994-01-01to2003-12-31_ext1.m", "MX"];
extTimeP = Import[directory <> "Model3.3_VCOutputs/model3.3.2_" <>
  species <> "_2014-01-01to2023-12-31_ext1.m", "MX"];
maxsims = 1000; γvals = Range[0, 2, .1]; tmax = 7300;
extMatrixH = ConstantArray[0, {Length[γvals], maxsims}];
Do[If[extTimeH[[i, l]] < tmax, extMatrixH[[i, l]] = 1, extMatrixH[[i, l]] = 0],
  {l, 1, maxsims}, {i, 1, Length[γvals]}]
propExtH = ConstantArray[0, {Length[γvals]}];
Do[propExtH[[i]] = Total[extMatrixH[[i]] / maxsims],
  {i, 1, Length[γvals]}]
propExtH = Transpose[{γvals, propExtH}];
extMatrixP = ConstantArray[0, {Length[γvals], maxsims}];
Do[If[extTimeP[[i, l]] < tmax, extMatrixP[[i, l]] = 1, extMatrixP[[i, l]] = 0],
  {l, 1, maxsims}, {i, 1, Length[γvals]}]
propExtP = ConstantArray[0, {Length[γvals]}];
Do[propExtP[[i]] = Total[extMatrixP[[i]] / maxsims],
  {i, 1, Length[γvals]}]
propExtP = Transpose[{γvals, propExtP}];

topl = {.22, .85}; bottomr = {.8, .2};
legendloc = bottomr; (*choose based on plot*)

```

```
fig3 = ListLinePlot[{propExtH, propExtP},
```



```

PlotStyle → {Gray, Black}, PlotRange → {{-0.02, 2.02}, {-0.02, 1.02}},
Frame → {True, True, False, False}, PlotLegends → Placed[
  {"Historic (1994–2003)", "Present (2014–2023)"}, legendloc], FrameStyle → 14,
FrameLabel → {"Spectral Exponent  $\gamma$ ", "Probability of Extinction"},
LabelStyle → Directive[Black],
Epilog → {Red, Dashed,
  Line[{{Pgamma, -1}, {Pgamma, Interpolation[propExtP][Pgamma]}]},
  Red, Dashed, Line[{{-1, Interpolation[propExtP][Pgamma]},
    {Pgamma, Interpolation[propExtP][Pgamma]}]},
  Lighter[Lighter[Red]], Dashed,
  Line[{{Hgamma, -1}, {Hgamma, Interpolation[propExtH][Hgamma]}]},
  Lighter[Lighter[Red]], Dashed, Line[{{-1, Interpolation[propExtH][Hgamma]},
    {Hgamma, Interpolation[propExtH][Hgamma]}]},
  {PointSize[0.015], Lighter[Lighter[Red]],
    Point[{Hgamma, Interpolation[propExtH][Hgamma]}]},
  {PointSize[0.015], Red, Point[{Pgamma, Interpolation[propExtP][Pgamma]}]}},
ImageSize → 500];

```

```
Print["Historical P(Ext) = ", Interpolation[propExtH][Hgamma]]
```

```
Print["Present P(Ext) = ", Interpolation[propExtP][Pgamma]]
```

```
(*combined=GraphicsGrid[{{fig1,fig2,fig3}},Spacings→{0,0},ImageSize→1500]*)
```

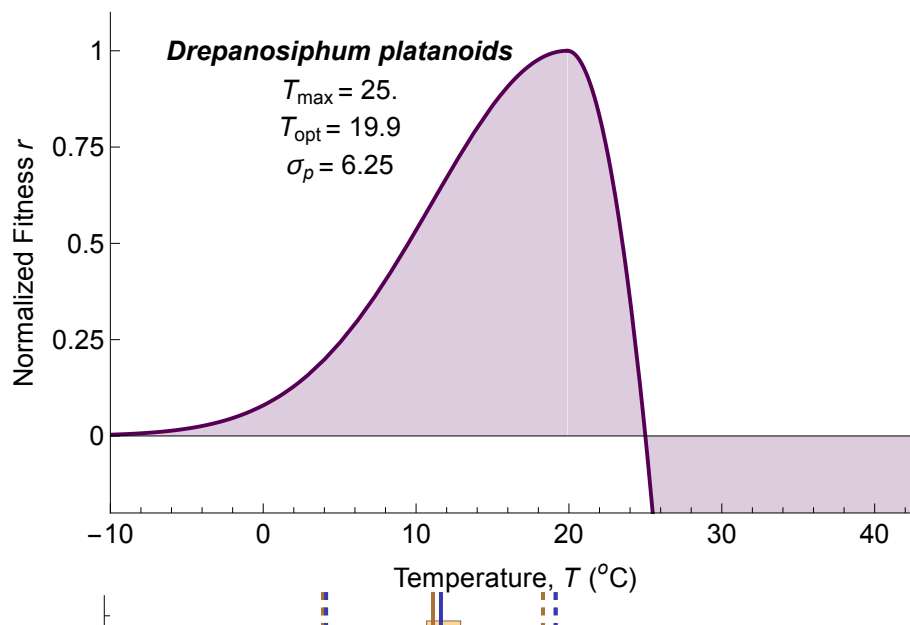
```
combinedpdf = GraphicsColumn[{{fig1, fig2, fig3}}, Spacings → {0, 0}, ImageSize → 500]
```

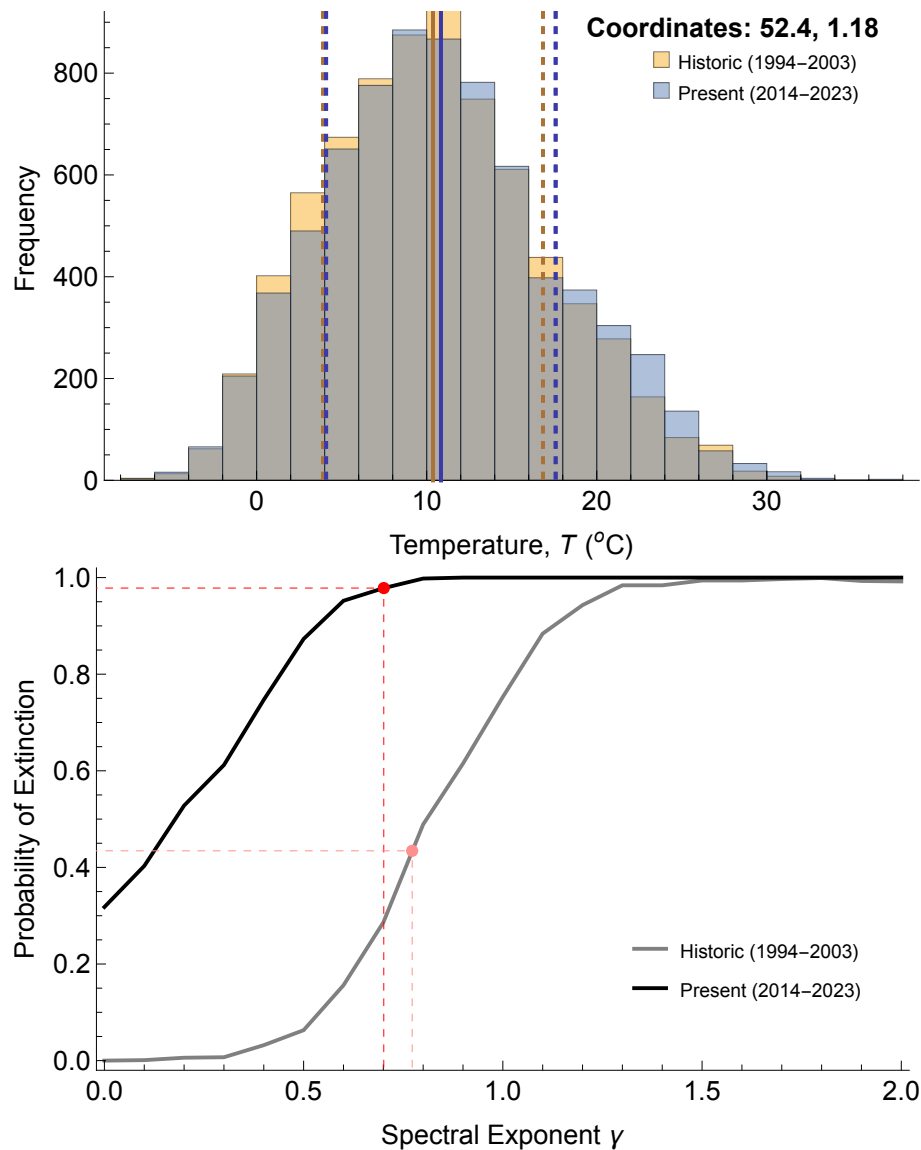
```
Export[saveto <> species <> "_all.pdf", combined, ImageResolution → 1000];
```

```
Historical P(Ext) = 0.434311
```

```
Present P(Ext) = 0.978292
```

Out[334]=





This code recreates plots of mean time to extinction for individual species that went extinct under 100% of scenarios.

```
In[94]:= getParameters[species_String] :=
  Module[{row}, row = SelectFirst[speciesData, #[[1]] == species &];
  If[row === Missing["NotFound"], "Species not found", Rest[row]]]
TPCdata = Import[directory <> "speciesparams.xlsx"];

species = "S.miscanthi";
extTimeH = Import[directory <> "Model3.3_VCOutputs/model3.3.2_" <>
  species <> "_1994-01-01to2003-12-31_ext1.m", "MX"];
extTimeP = Import[directory <> "Model3.3_VCOutputs/model3.3.2_" <>
  species <> "_2014-01-01to2023-12-31_ext1.m", "MX"];
speciesData = TPCdata[[1, 2 ;;]];
```

```

parameters = getParameters[species];
Hgamma = parameters[[6]];
Pgamma = parameters[[7]];
name = parameters[[14]];

γvals = Range[0, 2, .1];

extTimeMatrixH = ConstantArray[0, Length[γvals]];
Do[extTimeMatrixH[[i]] = Mean[extTimeH[[i]], {i, 1, Length[γvals]}]
extTimesH = Transpose[{γvals, extTimeMatrixH}];

extTimeMatrixP = ConstantArray[0, Length[γvals]];
Do[extTimeMatrixP[[i]] = Mean[extTimeP[[i]], {i, 1, Length[γvals]}]
extTimesP = Transpose[{γvals, extTimeMatrixP}];

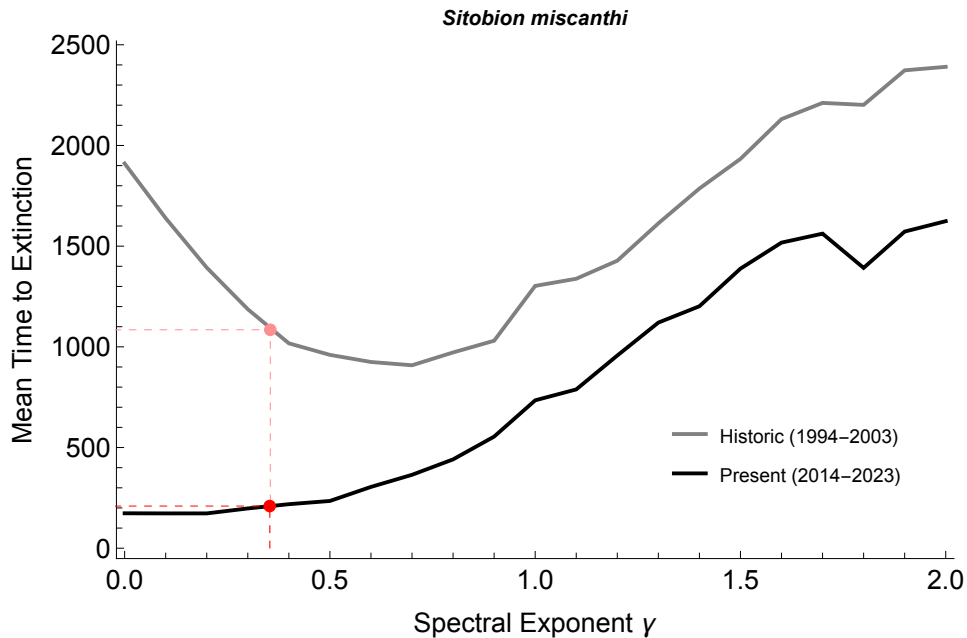
extTimePlot = ListLinePlot[{extTimesH, extTimesP},
  PlotStyle → {Gray, Black}, PlotRange → {{-0.02, 2.02}, Automatic},
  Frame → {True, True, False, False}, PlotLabel →
    Style[Text[Style[ToString[name], FontWeight → Bold, FontSlant → Italic]]],
  PlotLegends → Placed[{"Historic (1994–2003)", "Present (2014–2023)"}, {.8, .2}],
  FrameStyle → 14, FrameLabel → {"Spectral Exponent  $\gamma$ ", "Mean Time to Extinction"},
  LabelStyle → Directive[Black],
  Epilog → {Red, Dashed,
    Line[{Pgamma, -1}, {Pgamma, Interpolation[extTimesP][Pgamma]}],
    Red, Dashed, Line[{-1, Interpolation[extTimesP][Pgamma]},
      {Pgamma, Interpolation[extTimesP][Pgamma]}],
    Lighter[Lighter[Red]], Dashed,
    Line[{Hgamma, -1}, {Hgamma, Interpolation[extTimesH][Hgamma]}],
    Lighter[Lighter[Red]], Dashed, Line[{-1, Interpolation[extTimesH][Hgamma]},
      {Hgamma, Interpolation[extTimesH][Hgamma]}],
    {PointSize[0.015], Lighter[Lighter[Red]],
      Point[{Hgamma, Interpolation[extTimesH][Hgamma]}],
      {PointSize[0.015], Red, Point[{Pgamma, Interpolation[extTimesP][Pgamma]}]}],
    ImageSize → 500]

Print["Historical ExtTime = ", Interpolation[extTimesH][Hgamma]]
Print["Present ExtTime = ", Interpolation[extTimesP][Pgamma]]

Export[saveto <> "Model3.3_graphs/extTime/" <> species <> ".pdf",
  extTimePlot, ImageResolution → 1000];

```

Out[111]=



Historical ExtTime = 1084.68

Present ExtTime = 209.331

This code recreates plots of average minimum abundance for individual species that went extinct under 0% of scenarios.

In[115]:=

```
getParameters[species_String] :=
  Module[{row}, row = SelectFirst[speciesData, #[[1]] == species &];
  If[row === Missing["NotFound"], "Species not found", Rest[row]]]
TPCdata = Import[directory <> "speciesparams.xlsx"];

species = "B.argentifolia";
minPopH =
  Import[directory <> "Model3.3_VCOutputs/model3.3_minpop/model3.3.3_minPop_" <>
    species <> "_1994-01-01to2003-12-31_ext1.m", "MX"];
minPopP =
  Import[directory <> "Model3.3_VCOutputs/model3.3_minpop/model3.3.3_minPop_" <>
    species <> "_2014-01-01to2023-12-31_ext1.m", "MX"];
speciesData = TPCdata[[1, 2 ;;]];
parameters = getParameters[species];
Hgamma = parameters[[6]];
Pgamma = parameters[[7]];
name = parameters[[14]];

γvals = Range[0, 2, .1];
```

```

minPopMatrixH = ConstantArray[0, Length[γvals]];
Do[minPopMatrixH[[i]] = Mean[minPopH[[i]]], {i, 1, Length[γvals]}]
minPopsH = Transpose[{γvals, minPopMatrixH}];

minPopMatrixP = ConstantArray[0, Length[γvals]];
Do[minPopMatrixP[[i]] = Mean[minPopP[[i]]], {i, 1, Length[γvals]}]
minPopsP = Transpose[{γvals, minPopMatrixP}];

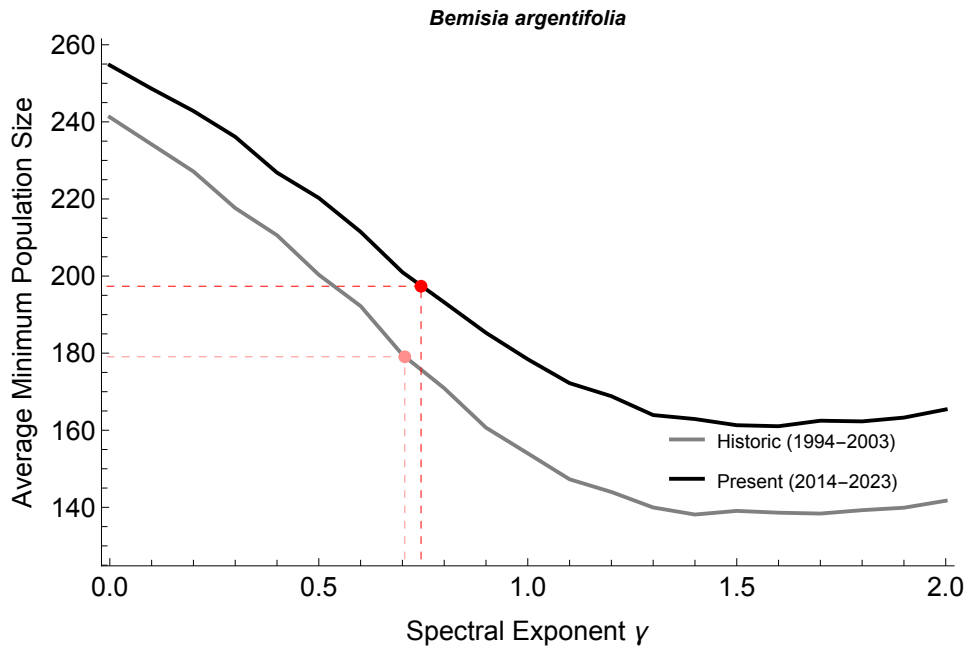
minPopPlot = ListLinePlot[{minPopsH, minPopsP},
  PlotStyle → {Gray, Black}, PlotRange → {{-0.02, 2.02}, Automatic},
  Frame → {True, True, False, False}, PlotLabel →
    Style[Text[Style[ToString[name], FontWeight → Bold, FontSlant → Italic]]],
  PlotLegends → Placed[{"Historic (1994-2003)", "Present (2014-2023)"}, {.8, .2}],
  FrameStyle → 14,
  FrameLabel → {"Spectral Exponent γ", "Average Minimum Population Size"},
  LabelStyle → Directive[Black],
  Epilog →
    {Red, Dashed, Line[{Pgamma, -1}, {Pgamma, Interpolation[minPopsP][Pgamma]}]},
    Red, Dashed, Line[{-1, Interpolation[minPopsP][Pgamma]},
      {Pgamma, Interpolation[minPopsP][Pgamma]}]},
    Lighter[Lighter[Red]], Dashed,
    Line[{Hgamma, -1}, {Hgamma, Interpolation[minPopsH][Hgamma]}]},
    Lighter[Lighter[Red]], Dashed, Line[{-1, Interpolation[minPopsH][Hgamma]},
      {Hgamma, Interpolation[minPopsH][Hgamma]}]},
    {PointSize[0.015], Lighter[Lighter[Red]],
      Point[{Hgamma, Interpolation[minPopsH][Hgamma]}]},
    {PointSize[0.015], Red, Point[{Pgamma, Interpolation[minPopsP][Pgamma]}]}},
  ImageSize → 500]

Print["Historical Minimum Pop Size = ", Interpolation[minPopsH][Hgamma]]
Print["Present Minimum Pop Size = ", Interpolation[minPopsP][Pgamma]]

Export[saveto <> "Model3.3_graphs/minPop/" <> species <> ".pdf",
  minPopPlot, ImageResolution → 1000];

```

Out[132]=



Historical Minimum Pop Size = 179.056

Present Minimum Pop Size = 197.359

This code allows you to extract the spectral exponent and, if desired, plot the spectral density of any of the temperature time series.

In[136]:=

```

getParameters[species_String] :=
Module[{row}, row = SelectFirst[speciesData, #[[1]] == species &];
If[row == Missing["NotFound"], "Species not found", Rest[row]]]

TPCdata = Import[directory <> "speciesparams.xlsx"];
speciesData = TPCdata[[1, 2 ;;]];
species = "C.shadabi";
historic = "1994-01-01to2003-12-31";
present = "2014-01-01to2023-12-31";

parameters = getParameters[species];
lat = ToString[parameters[[4]]];
long = ToString[parameters[[5]]];

data = Import[directory <> "VisualCrossingData/" <> species <>
  "_" <> lat <> "," <> long <> "_" <> historic <> ".csv", "CSV"];
temps = Flatten[data[[2 ;;, {2, 3}]]];
lt = Flatten[Abs[Fourier[{temps}]]^2];
lt = Drop[lt, -Length[lt] / 2];
lt = Drop[lt, 1];
lt0 = MapIndexed[{#2[[1]], #1} &, lt];
fit0 = Fit[Log[lt0], {1, x}, x]

data = Import[directory <> "VisualCrossingData/" <> species <>
  "_" <> lat <> "," <> long <> "_" <> present <> ".csv", "CSV"];
temps = Flatten[data[[2 ;;, {2, 3}]]];
lt = Flatten[Abs[Fourier[{temps}]]^2];
lt = Drop[lt, -Length[lt] / 2];
lt = Drop[lt, 1];
lt0 = MapIndexed[{#2[[1]], #1} &, lt];
fit0 = Fit[Log[lt0], {1, x}, x]

Show[ListLinePlot[lt, ScalingFunctions -> {"Log", "Log"},
  Frame -> True, FrameLabel -> {"frequency", "spectral density"}],
  Plot[2.6340898511613706 - 0.3970554074582351 x, {x, 0, tmax}, PlotStyle -> Red]]

```

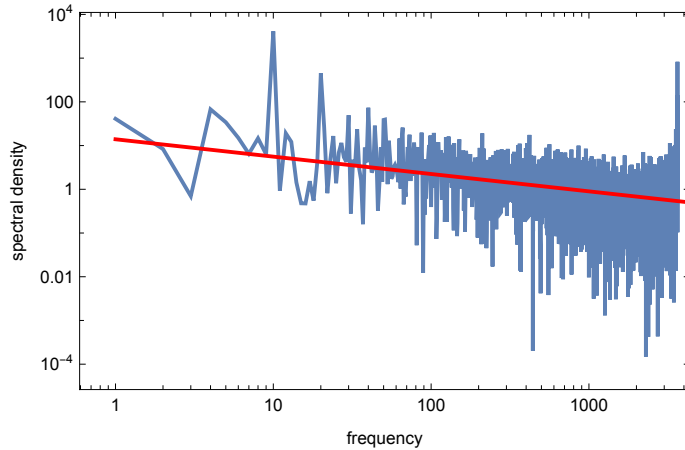
Out[151]=

2.63409 - 0.397055 x

Out[158]=

2.27987 - 0.36167 x

Out[159]=



Bar chart summary of results

This code recreates Fig 5c and d.

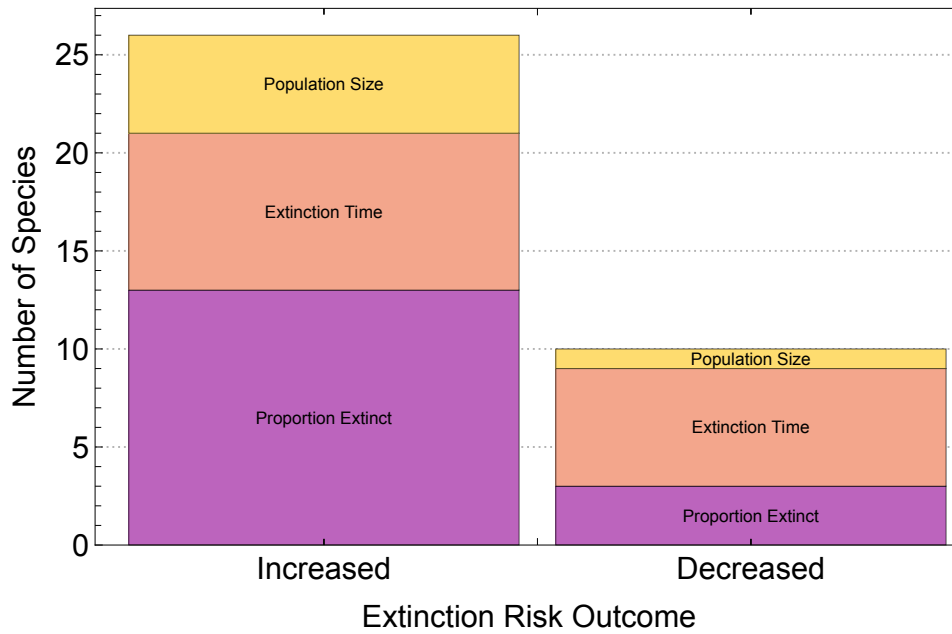
In[160]:=

```
outcomebars = BarChart[{{13, 8, 5}, {3, 6, 1}},
  ChartLayout → "Stacked", ChartStyle → {Lighter[RGBColor["#9a169c"]],
    Lighter[RGBColor["#ed7953"]], Lighter[RGBColor["#fdca27"]]},
  ChartLabels → {"Increased", "Decreased"},
    {"Proportion Extinct", "Extinction Time", "Population Size"},
  PlotTheme → "Detailed", ImageSize → 500,
  FrameLabel → {"Extinction Risk Outcome", "Number of Species"},
  FrameStyle → 16, "FixedBarSpacing" → True]

regionbars = BarChart[{{17, 8}, {3, 2}, {6, 0, 2}}, ChartLayout → "Stacked",
  ChartStyle → {RGBColor["#c0665e"], RGBColor["#aeceeb"], RGBColor["#dacfbe"]},
  ChartLabels → {"Northern\n Hemisphere", "Tropics", "Southern\n Hemisphere"},
    {"Higher Risk", "Lower Risk", "No Change"}, PlotTheme → "Detailed",
  ImageSize → 500, FrameLabel → {None, "Number of Species"},
  FrameStyle → 16, "FixedBarSpacing" → True]

Export[saveto <> "barchart_outcome.pdf", outcomebars, ImageResolution → 1000];
Export[saveto <> "barchart_region.pdf", regionbars, ImageResolution → 1000];
```


Out[160]=



Out[161]=

